

Image transformation

Intensity Transformation: $s = T(r)$

output input

① Threshold

$$s = \begin{cases} 0 & r < T \\ 1 & r \geq T \end{cases}$$

② Negative $\rightarrow L = 2^k$, intensity $s \in [0, L-1]$

$$s = L - 1 - r$$

"Enhancing white or grey detail"

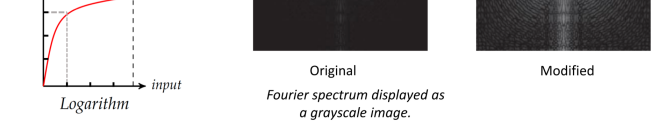
③ Log transformation

$$s = c \log(1+r)$$

- Maps a narrow range of low intensity values in the input into a wider range of output levels.
- Stretch low intensity levels, while compressing the high intensity levels.

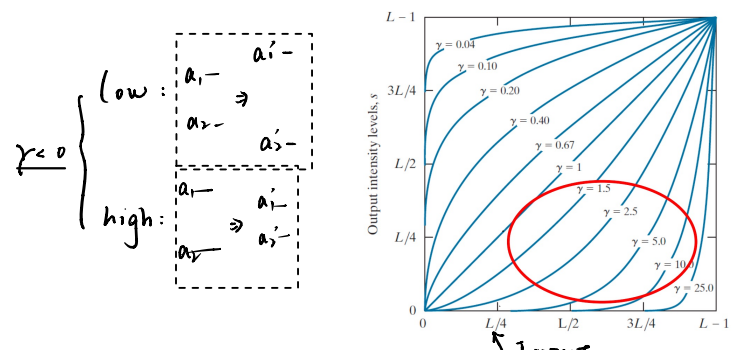
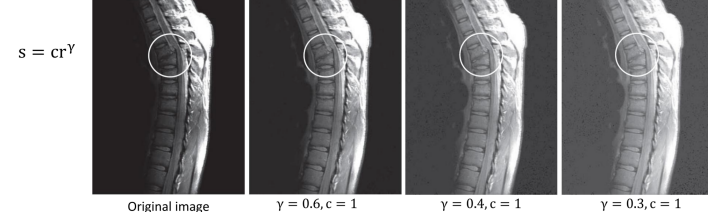
$$s = c \log(1+r)$$

where c is a constant.

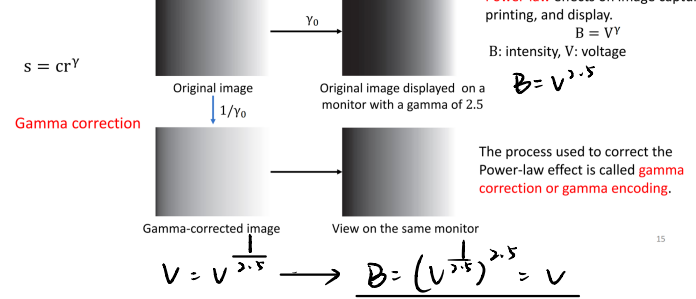


Power-law (Gamma) Transformations

- Some basic intensity transformation functions: **Power-law transformations**
- For $\gamma < 1$, the transformation will stretch low intensity levels and compress high intensity levels.
- PRESERVE details in the low intensity levels. Wash-out appearance at a small gamma value.

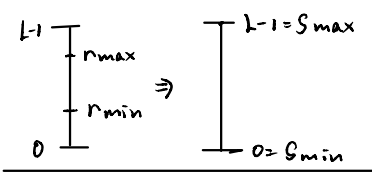


- Applications: Gamma correction



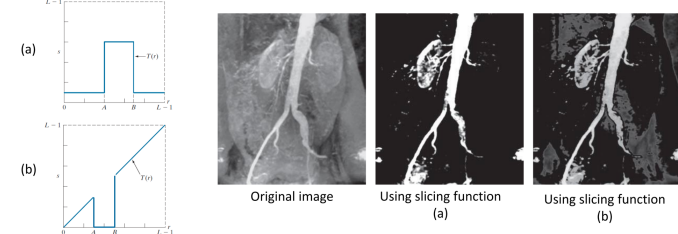
Contrast stretching

$$s = \frac{r - r_{\min}}{r_{\max} - r_{\min}} (L - 1)$$

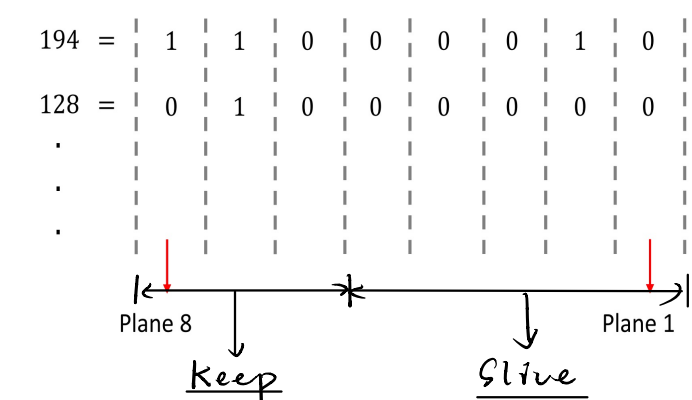


Intensity-level Slicing

- Some basic intensity transformation functions: **Piece-wise linear transformation function**
- Intensity-level slicing: highlight a specific range of intensities

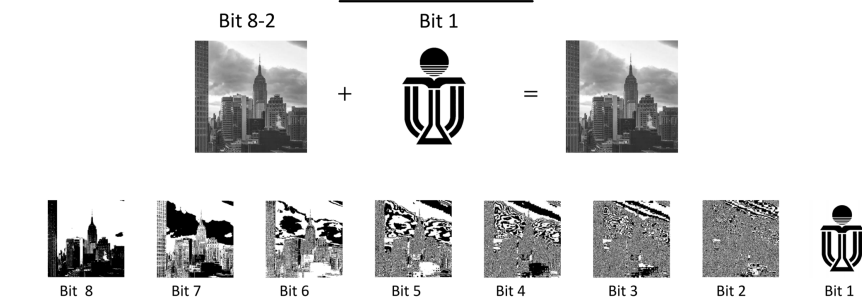


Bit-plane Slicing



Can lower storage $\Rightarrow$ image compression

Bit-plane slicing application in digital watermarking: hiding information digitally



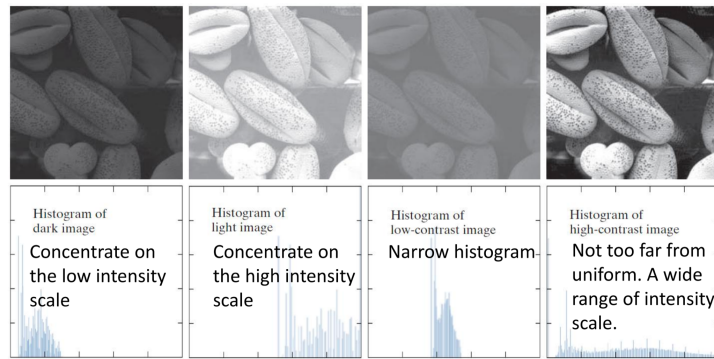
Histogram Processing

Histogram: distribution of all pixel values

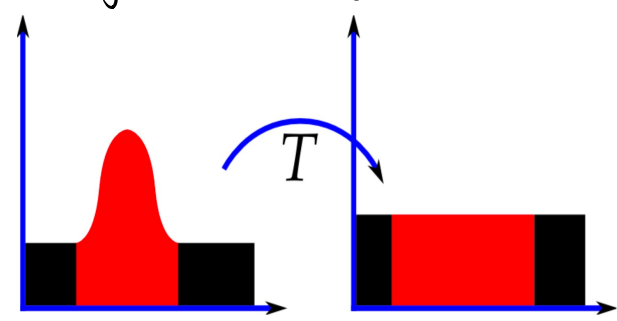
$$\Rightarrow h(r_k) = n_k = \# \text{ pixels with intensity } r_k$$

Normalized Histogram:

$$P(r_k) = \frac{n_k}{MN} \in [0, 1]$$



Histogram Processing



What we want: "equally" many pixels at every gray level

$\Rightarrow P(r_k)$ approximately follows an uniform distribution

$$s = T(r) \quad T(r) \text{ should specify: } 1. T(r) \geq T(r') \text{ for } r > r' \quad 2. 0 \leq T(r) \leq L-1 \text{ for } 0 \leq r \leq L-1$$

$$\text{cdf: } F_S(s) = P(S \leq s) = P(T(R) \leq s) = P(R \leq T^{-1}(s)) = F_R(T^{-1}(s))$$

$$\Rightarrow \frac{dF_S(s)}{ds} = \frac{dF_R(T^{-1}(s))}{ds} \quad \frac{d}{ds} T^{-1}(s) = \frac{1}{T'(T^{-1}(s))}$$

$$P_S(s) = P_R(T^{-1}(s)) \cdot \frac{d}{ds} T^{-1}(s)$$

$$P_S(s) = P_R(T^{-1}(s)) \cdot \frac{1}{T'(T^{-1}(s))}$$

$$P_S(s) = P_R(r) \cdot \frac{1}{T'(r)} = P_R(r) \cdot \left| \frac{dr}{ds} \right|$$

$$\Rightarrow P_S(s) = P_R(r) \cdot \left| \frac{dr}{ds} \right|$$

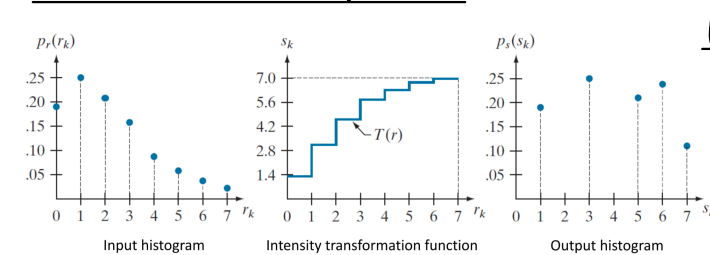
$$\text{Assume } T(r) = (L-1) \int_0^r P_R(u) du \quad \text{"cdf: } F_R(r) \text{"}$$

$$\Rightarrow \frac{ds}{dr} = (L-1) \cdot P_R(r)$$

$$P_S(s) = P_R(r) \cdot \frac{1}{(L-1) P_R(r)} = \frac{1}{L-1} \Rightarrow \text{"Uniform distribution"}$$

In discrete case:

$$T(r_k) = (L-1) \sum_{i=0}^k P_R(r_i)$$



We can only approximately achieve uniform distribution

Is histogram equalization always good?

No! It may increase the contrast of background noise, while decreasing the useful signal or it can remove small details.

Histogram matching



$P_R(r)$: pdf of r .

$P_S(s)$: specified pdf

① r -equalization $\rightarrow s$.

$$s = T(r) = (L-1) \int_0^r P_R(u) du$$

② z -equalization $\rightarrow G(z)$

$$G(z) = (L-1) \int_0^z P_S(u) du$$

③ Let: $s = G(z)$

$$T(r) = G(z) \quad z = G^{-1}(T(r))$$

$$\text{Discrete: } s_k = T(r_k) = \frac{(L-1)}{MN} \sum_{j=0}^k n_j$$

$$G(z_k) = (L-1) \sum_{i=0}^k P_S(z_i) = s_k$$

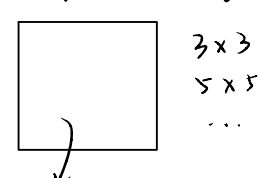
$$z_k = G^{-1}(s_k)$$

Histogram Processing

Histogram Equalization: global. Cannot enhance local detail.

Local Histogram Equalization

1. Define a neighborhood



2. Histogram Eq in this neighbourhood

Only update centered pixel

Basic Elements Arithmetic Operation

Elementwise Product

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \odot \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

Matrix Product

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

if $H[f_1(x,y) + f_2(x,y)] = aH[f_1(x,y)] + bH[f_2(x,y)]$

then $H()$ is linear operator

$$s(x,y) = f(x,y) + g(x,y)$$

$$d(x,y) = f(x,y) - g(x,y)$$

$$p(x,y) = f(x,y) \cdot g(x,y)$$

$$v(x,y) = f(x,y) \div g(x,y)$$

Addition: Denoise

image with noise: sample n times.

with each: $f_i(x,y) = f(x,y) + n_i(x,y)$

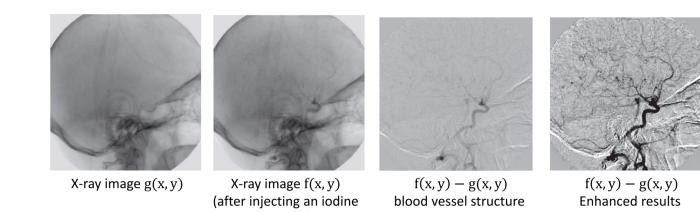
$$\bar{f}(x,y) = \frac{1}{n} \sum_{i=1}^n f_i(x,y) = f(x,y) + \frac{1}{n} \sum_{i=1}^n n_i(x,y)$$

$$\Rightarrow E[\bar{f}(x,y)] = f(x,y) \quad \sigma_{\bar{f}(x,y)} = \frac{1}{\sqrt{n}} \sigma_{n(x,y)}$$

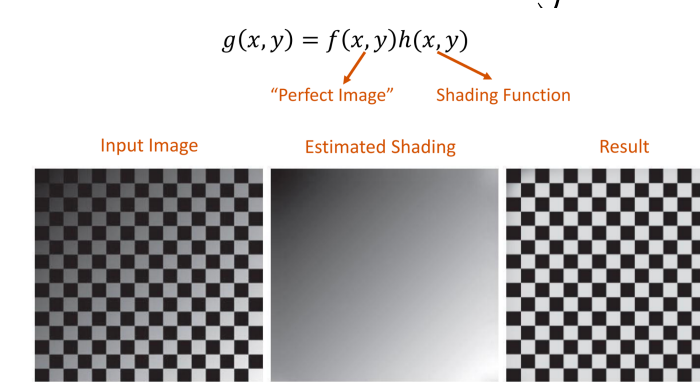
Subtraction: enhance difference

Image Subtraction

- Image subtraction to enhance difference for comparison.
- Application: Mask mode radiography



Multiplication/Division: Shading Correction



deshading: $f(x,y) = \frac{g(x,y)}{h(x,y)}$

Image Multiplication/Division for Masking

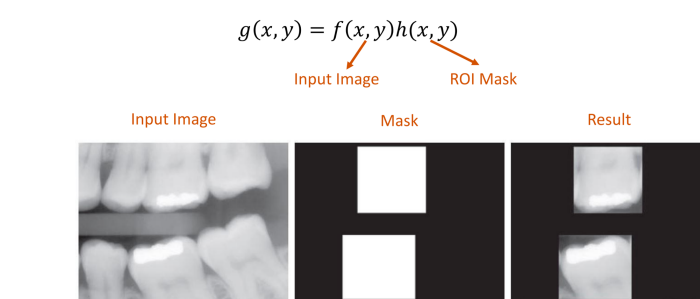


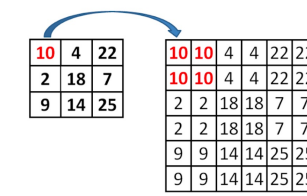
Image Interpolation

$\Rightarrow$ Change the coordinate

Key: Figure out how pixels in output related to one or group of pixels in input.

Nearest Neighbor

Assign to each new location the intensity of its nearest neighbor in the original image



Bilinear

$$\begin{aligned} \text{① } x\text{-coordinate:} \\ f(x,y) &= \frac{x_2 - x}{x_2 - x_1} f(Q_{21}) + \frac{x - x_1}{x_2 - x_1} f(Q_{11}) \\ f(x,y) &= \frac{x_2 - x}{x_2 - x_1} f(Q_{22}) + \frac{x - x_1}{x_2 - x_1} f(Q_{12}) \\ \text{② } y\text{-coordinate:} \\ f(x,y) &= \frac{y - y_1}{y_2 - y_1} f(x, y_2) + \frac{y_2 - y}{y_2 - y_1} f(x, y_1) \end{aligned}$$

Bicubic: similar to bilinear, better

Affine transformation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Transformation Name	Affine Matrix, A	Coordinate Equations	Example	Translation	$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} x' = x + t_x \\ y' = y + t_y \end{bmatrix}$	$\begin{bmatrix} T \\ T \\ T \end{bmatrix}$
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = y$					
Scaling/Reflection (For reflection, set one scaling factor to -1 and the other to 0)	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = c_x x$ $y' = c_y y$					
Rotation (about the origin)	$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x \cos \theta - y \sin \theta$ $y' = x \sin \theta + y \cos \theta$					
Shear (vertical)	$\begin{bmatrix} 1 & s & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + sy$ $y' = y$					
Shear (horizontal)	$\begin{bmatrix} 1 & 0 & 0 \\ s & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + sy$ $y' = y$					

